

# Sai Yasaswini Majety

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**Work Authorization: Approved H-4 EAD (Unrestricted U.S. Work Authorization : No Visa Sponsorship Needed)**

## Professional Summary

Senior Backend and Cloud Infrastructure Engineer with **6+ years of production experience** designing, deploying, and hardening high-availability distributed systems, real-time fleet telemetry pipelines, and enterprise developer platforms. Proven engineering track record at **Viasat** architecting mission-critical ground-to-air microservices, executing zero-downtime AWS database migrations across Multi-AZ RDS clusters, and engineering automated TLS and OAuth 2.0 enterprise security gateways. Co-inventor on published Indian Patent **202541026299** for modular deep learning architectures with parameter-efficient adapter routing. Deep experience building resilient event-driven architectures, automated regression test harnesses, and regulated compliance audit trails directly aligned with enterprise healthcare and dose management platform requirements. Authorized to work for any U.S. employer without visa sponsorship.

## Technical Skills

**Backend & Microservices:** Python (FastAPI, Flask, Django), Go, RESTful APIs, gRPC, Event-Driven Architecture, Microservices Design, Asynchronous Programming, High-Throughput Request Routing  
**Cloud Infrastructure & DevOps:** AWS (ECS, RDS Aurora/PostgreSQL/MySQL, CloudFormation, S3, IAM, CloudWatch, KMS), Docker, Kubernetes, Jenkins CI/CD, Terraform / IaC, Linux/Unix Systems Administration  
**Security & Identity Governance:** OAuth 2.0, Okta, SAML 2.0, Automated TLS/SSL X.509 Certificate Lifecycle, HashiCorp Vault, AWS Secrets Manager, Least-Privilege IAM Access Controls, 21 CFR Part 11 Audit Trail Integrity  
**Databases & Distributed Caching:** AWS RDS PostgreSQL, MySQL Multi-AZ, MariaDB, Redis In-Memory Cache, Flyway-Style Idempotent Schema Migrations, Sphinx Full-Text Search  
**Telemetry & Streaming Systems:** Real-Time Fleet Telemetry Ingestion, Store-and-Forward Edge Queues, Message Brokers (Kafka, MQTT, Kinesis Patterns), Health Heartbeats, Backpressure Mitigation  
**Testing & Observability:** Pytest, Robot Framework, Postman API Automation, End-to-End Regression Suites, CloudWatch Alarms and Dashboards, Synthetic Endpoint Monitoring  
**AI Platforms & Data Engineering:** Retrieval-Augmented Generation (RAG), Parameter-Efficient Adapter Routing, FAISS Vector Indexing, Hugging Face, LangChain, Streamlit

## Professional Experience

### Viasat Inc. / Viasat India Private Limited

Chennai, India

Senior Software Engineer - I (Backend & Cloud Infrastructure)

Jan 2023 - Feb 2024

- **Commercial Aviation Network Access & Fleet Telemetry Platform:** Core ground-to-air connectivity platform delivering wireless access, passenger portals, and real-time fleet telemetry across **1,000+ commercial aircraft** (including Delta Air Lines fleets).
- Scaled real-time passenger portal and telemetry API microservices to **99.99% uptime across 1,000+ commercial aircraft** by engineering asynchronous Python FastAPI gateways and distributed Redis caching.
- Eliminated customer downtime during multi-terabyte transactional database upgrades by executing **zero-downtime live schema migrations** across Multi-AZ AWS RDS MySQL clusters (v5.6 to 5.7 and 8.0).
- Automated enterprise TLS/SSL certificate lifecycle across **500+ edge gateways and endpoints**, eliminating manual renewals and preventing outage vulnerabilities (**Viasat Spotlight Award**).
- Designed decoupled OAuth 2.0 and SAML/Okta identity gateway routing, reducing third-party partner integration onboarding time from **3 weeks to 2 days (Viasat Architecture Commendation)**.
- Hardened in-flight fleet connectivity across intermittent satellite links by implementing exponential backoff retry algorithms, store-and-forward edge queues, and synthetic health-check heartbeats.
- Enforced strict audit trail logging and configuration state governance across microservices, ensuring traceability for high-assurance aviation service compliance.

### Viasat Inc. / Viasat India Private Limited

Chennai, India

Software Engineer II (Cloud Platform & API Microservices)

Jan 2021 - Dec 2022

- Accelerated multi-region cloud deployment velocity by **65% (from 4 hours to 45 minutes MTTR)** by authoring modular AWS CloudFormation IaC templates and declarative Jenkins CI/CD pipelines.
- Elevated test coverage across critical billing, telemetry, and authentication paths from **60% to >92%** by architecting automated Pytest suites and mock network failure harnesses.
- Engineered automated database backup validation, failover health checks, and cross-region replication monitoring across enterprise AWS RDS instances.

- **Commercial Air PAT (Pipeline & Telemetry Observability):** Enterprise real-time observability platform tracking test execution velocity, release health, and software reliability across global engineering groups.
- Reduced release cycle verification latency by **40%** by engineering a real-time observability dashboard using Python Flask and React, aggregating telemetry across 50+ distributed Jenkins nodes.
- Standardized release quality gates by programming automated pre-merge regression verification pipelines in Groovy, preventing breaking API regressions from reaching staging environments.
- Authored comprehensive regression test suites spanning frontend UI and backend handshakes using **Robot Framework**, executing **100+ automated test scenarios per release candidate**.
- **CandEx 2.0 Platform (Internal Talent Operations & Search Engine):** Enterprise talent operations and candidate management platform streamlining high-volume hiring workflows.
- Accelerated technical resume search query response times from **3.2s to <180ms** by integrating Sphinx full-text search engine with Python Django microservices across tens of thousands of candidate profiles.
- Optimized relational data models and indexed schemas across MariaDB instances, supporting concurrent multi-recruiter workflows with zero database locking contention.

## Patents & Research Contributions

### Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning

Indian Patent Office (IPO) : Application No. 202541026299 (Published in The Patent Office Journal No. 13/2025)

Co-inventor of a parameter-efficient adapter architecture enabling deep neural networks to dynamically route representations across disparate cross-domain distributions without catastrophic forgetting. Engineered dynamic adapter routing layers and empirical evaluation benchmarks.

## Open-Source Systems & Applied AI Projects

### RAG Document Intelligence Assistant (rag-hf-chatbot)

GitHub Open Source

Core Systems Engineer (Python, Hugging Face, FAISS, Streamlit)

2024

- Engineered an interactive document question-answering system using Hugging Face transformer embeddings and FAISS vector indexing, achieving sub-200ms semantic context retrieval over complex technical PDF documentation.
- Implemented chunking and embedding caching pipelines to minimize inference overhead while preserving syntactic context boundaries.

### ChatGPT Contextual Tool Execution Assistant

GitHub Open Source

Core Systems Engineer (Python, OpenAI API, LangChain)

2024

- Architected a modular conversational agent integrating external API tool-calling capabilities with structured memory state persistence.

## Certifications & Technical Honors

- **Viasat Spotlight Award (2023):** Automated TLS/SSL Edge Security and X.509 Certificate Infrastructure Lifecycle
- **Viasat Architecture Commendation (2022):** Architecture and Delivery of OAuth 2.0 & Enterprise Identity Gateway
- **Viasat Engineering Excellence (2022):** Zero-Downtime Multi-AZ Cloud Database Migration and Clustering
- **1st Place Winner (2019):** Viasat IoT & Cloud Telemetry Enterprise Hackathon
- **Dean's Merit Scholarship (2016 - 2017):** Top 3% Academic Distinction, SASTRA Deemed University

## Education

### SASTRA Deemed University

Thanjavur, India

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering (EEE)

Aug 2014 - May 2018

- **Academic Distinction:** Cumulative GPA: **8.2 / 10.0** | Recipient of **Dean's Merit Scholarship** (Top 3% cohort ranking).
- **Core Coursework:** Microprocessor & Microcontroller Architectures, Control Systems, Data Structures, OOP (Python, C++), Relational Databases & SQL.
- **Campus Leadership:** Technical Infrastructure & Workshop Organizer, DAKSH National Techno-Management Fest; Peer Programming Mentor.